

- (b) Which observations led to the conceptualization of neutral theory of molecular evolution? Discuss the salient features of the proposed theory. (6)
5. (a) Describe and exemplify the types of speciation initiated by extrinsic and intrinsic isolation. (8)
- (b) Discuss briefly the type of mass extinction resulting from asteroid impact. (4)
6. (a) Enumerate the trends in the evolutionary progression of horse phylogeny with reference to important fossil records. (8)
- (b) How do you infer phylogenetic tree topology? (4)
7. Write short notes on **any three** of the following : (12)
- (a) Darwin's theory and its demerits
- (b) RNA world
- (c) Adaptive radiation
- (d) Types of Natural Selection
- (e) Unique hominin characteristics

(1000)

[This question paper contains 4 printed pages.]

31.05.2024(M)
Your Roll No.....

Sr. No. of Question Paper : 2992

Unique Paper Code : 32231602

Name of the Paper : Evolutionary Biology

Name of the Course : B.Sc. (Hons.) Zoology
(LOCF)

Semester : VI

Duration : 3 Hours

Maximum Marks : 75

Instructions for Candidates

- Write your Roll No. on the top immediately on receipt of this question paper.
- Draw well-labeled diagrams wherever necessary.
- Attempt **five** questions in all. Question No. 1 is compulsory.

- (a) Define the following : (5)
 - Gene pool
 - Genetic load
 - Deme
 - Orthologous genes
 - Haldane's rule

P.T.O.

(b) Differentiate between the following : (10)

- (i) Cladogram and Phylogram
- (ii) Anagenesis and Cladogenesis
- (iii) Species and Subspecies
- (iv) Bottle neck effect and Founder effect
- (v) Homologous and Analogous organs

(c) Give the major contribution of the following in the field of Evolutionary biology : (5)

- (i) Georges Cuvier
- (ii) Gould and Eldredge
- (iii) Lynn Margulis
- (iv) HBD Kettlewell
- (v) Donald Johanson

(d) Match the following : (3)

Column A	Column B
(i) Stromatolites	A. Migration
(ii) Circle of races	B. Land tetrapods
(iii) Trace fossils	C. Inclusive fitness
(iv) Devonian period	D. Filamentous Cyanobacteria
(v) Gene flow	E. Coprolites
(vi) Kin selection	F. Ressenkries

(e) Justify the following statements : (4)

- (i) Fragmentation of habitats result in speciation on one hand and extinction on the other.
- (ii) Both Lamarck's and Darwin's theory can be explained by the example of giraffe's neck.

2. (a) What are the assumptions of Hardy Weinberg equilibrium? Add a note on the implications and significance of the principle as applied to natural populations. (6)

(b) Describe the various pre-zygotic isolating mechanism with suitable examples. (6)

3. (a) How do fossils provide direct evidence for evolution? Briefly describe the various dating methods for fossils. (6)

(b) Organic variations provide raw material for evolution. Support the statement with the help of examples. (6)

4. (a) Explain how simple chemical compounds in the early environment gave rise to basic bio molecules. (6)